

Wrapping up the Python Crash Course

IDSP Lecture 11

September 30, 2025

Plan for today

Part 1

- * Coding test details
- * Promotion of Lecture 10 (e-learning unit on webscraping)
- * ``conda`` (PS from Lecture 09)
- * Next steps for your programming career (some propaganda from Anastassia)

Part 2

- * Big recap:
 - (dictionaries);
 - control flow;
 - while loops;
 - pandas;
 - random module

Coding test details

- * **16:00** Be there
- * **16:15** Test is released on learnIT at
- * **18:00** Submission deadline
- * ... questions?

Lecture 10 e-learning unit promotion

- * What is HTML?
- * ``requests`` library for sending HTTP requests
- * ``beautifulsoup4`` for parsing HTTP responses

▼ Lecture 10 (Sep 25)

Exercises (14-16h) and Study Lab (16-18h) take place in person as usual.

No on-site lecture - e-learning session instead!

Topics covered in the e-learning session: Webscraping with the beautifulsoup4 package (see CM book, Chapter 10 Thursday & AS book Chapter 13)

Topics covered in the exercises: coding test preparation.



e-learning session: webscraping



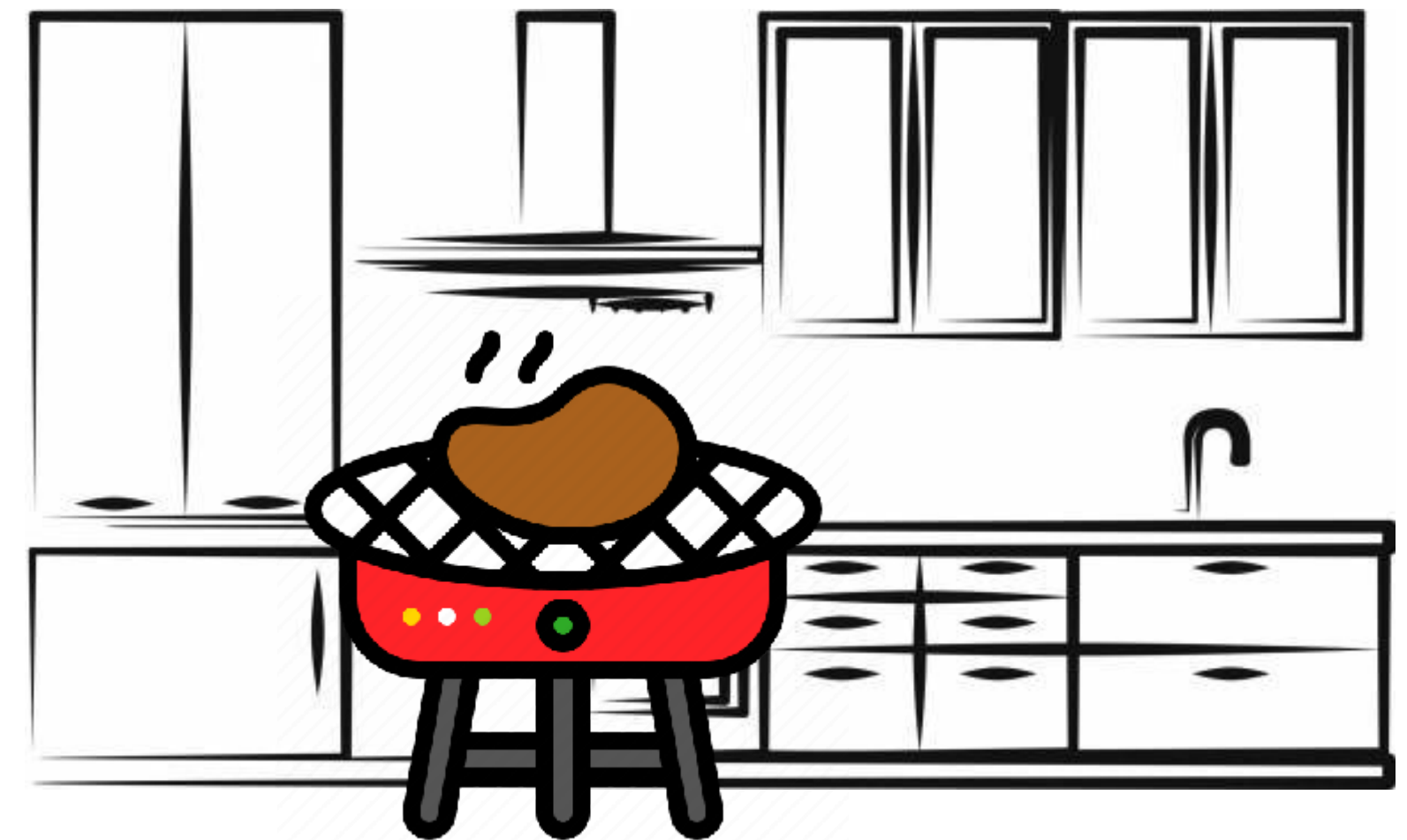
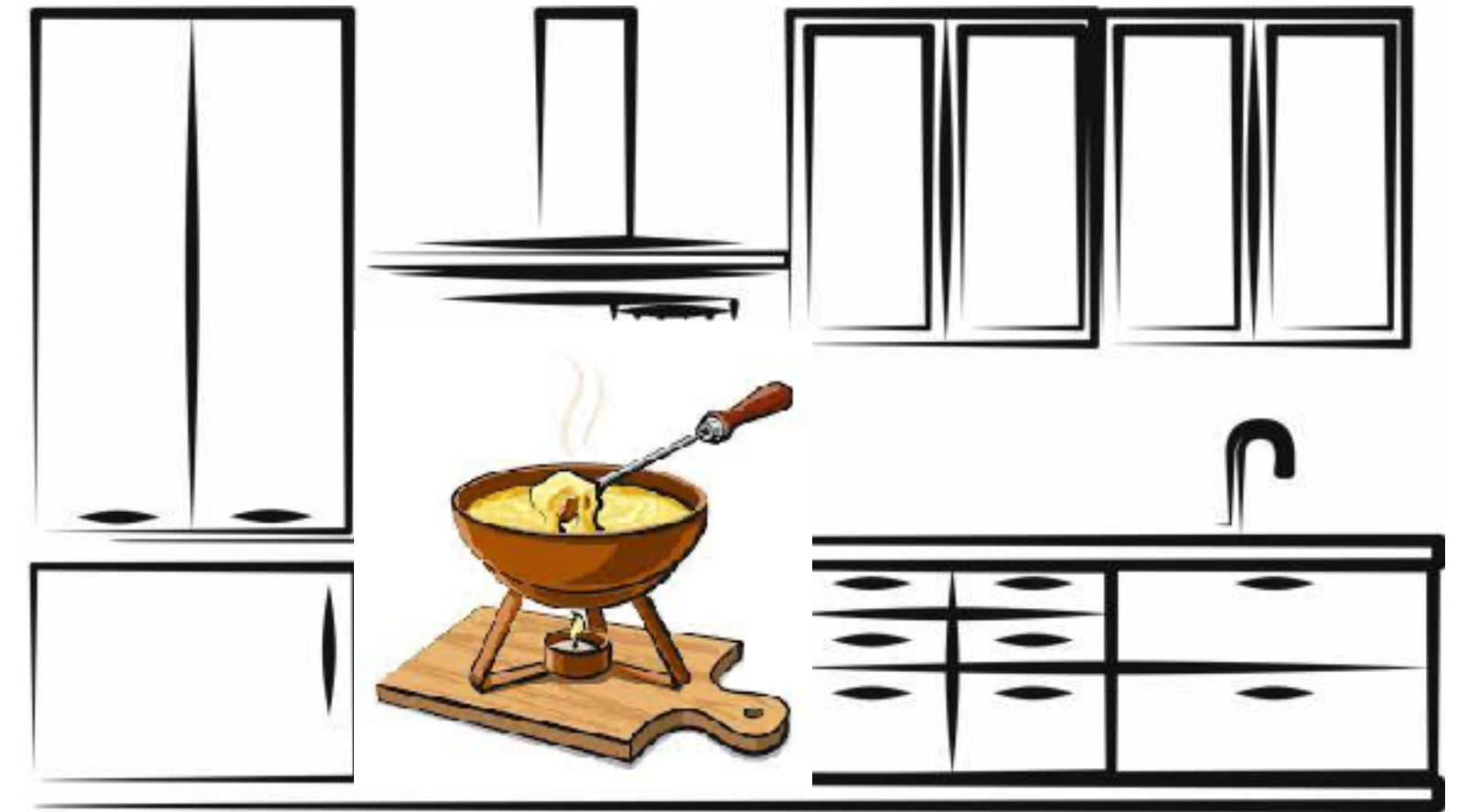
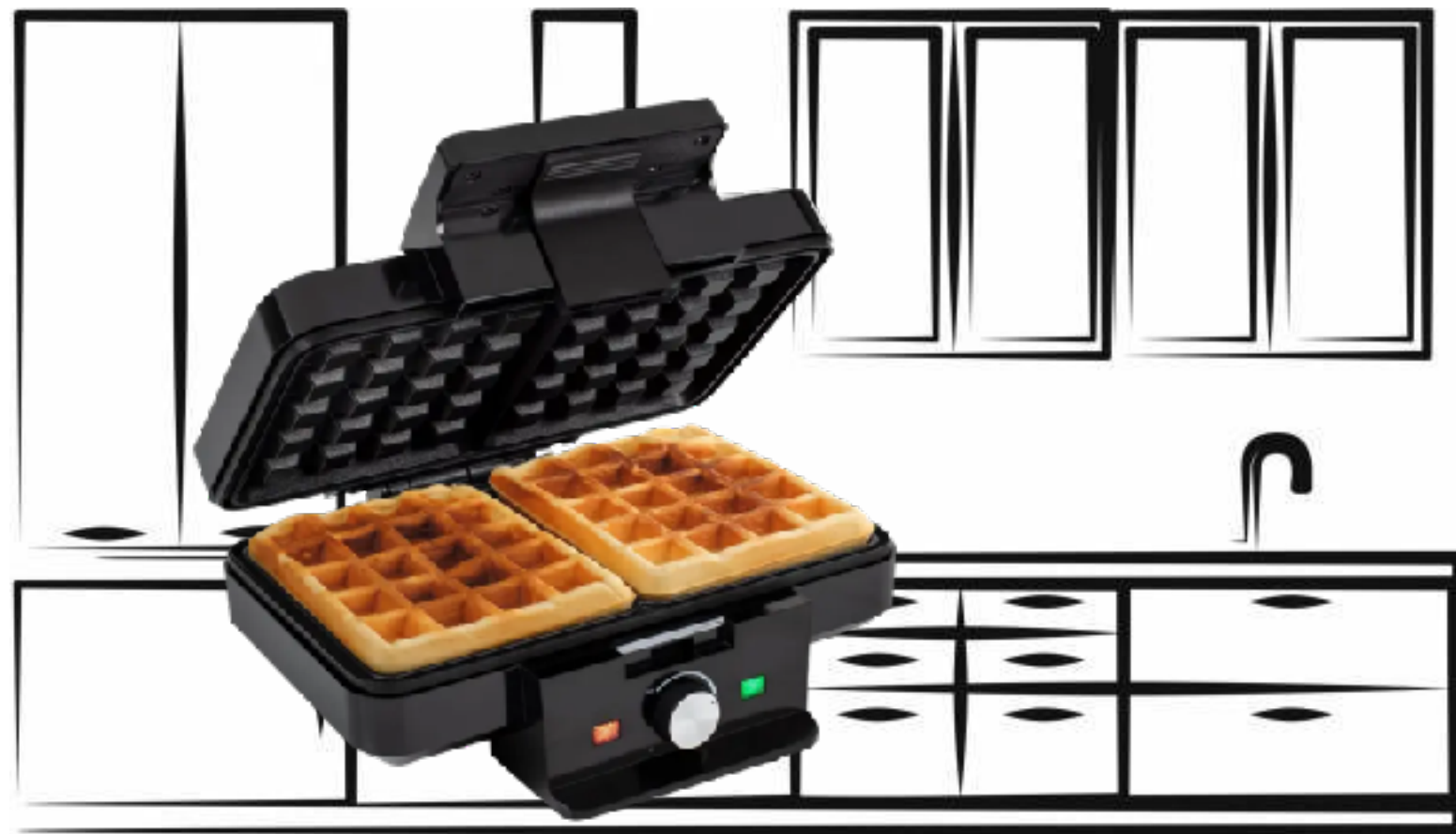
e-learning recoding (webscraping)

PS to Lecture 09 - Managing environments with conda

Introduction to Data Science and Programming



Every project has its own **ENVIRONMENT**



Environments with conda

From the command line

- `conda list` **List all packages in CURRENT environment**
- `conda env list` **List all environments**
- `conda create --name <environmentname>` **Create a new environment**
- `conda create --name <environmentname> <packagename>` **Create a new environment + packages**
- `conda activate <environmentname>` **“Enter” the new environment**
- `conda install <packagename>` **Install a new package in the CURRENT environment**
- `conda deactivate` **“Get out of” the new environment (back to base)**
- `conda remove --name <environmentname> --all` **Remove an environment**

Environments with conda...

... where Jupyter Notebook works

✓ jupyter and ipykernel installed

✓ kernelspec installed:

```
python -m ipykernel install --user --name <environmentname>
```

✓ Select kernel to run notebook (shown in class: in browser & in VS Code)

Try it out yourself!

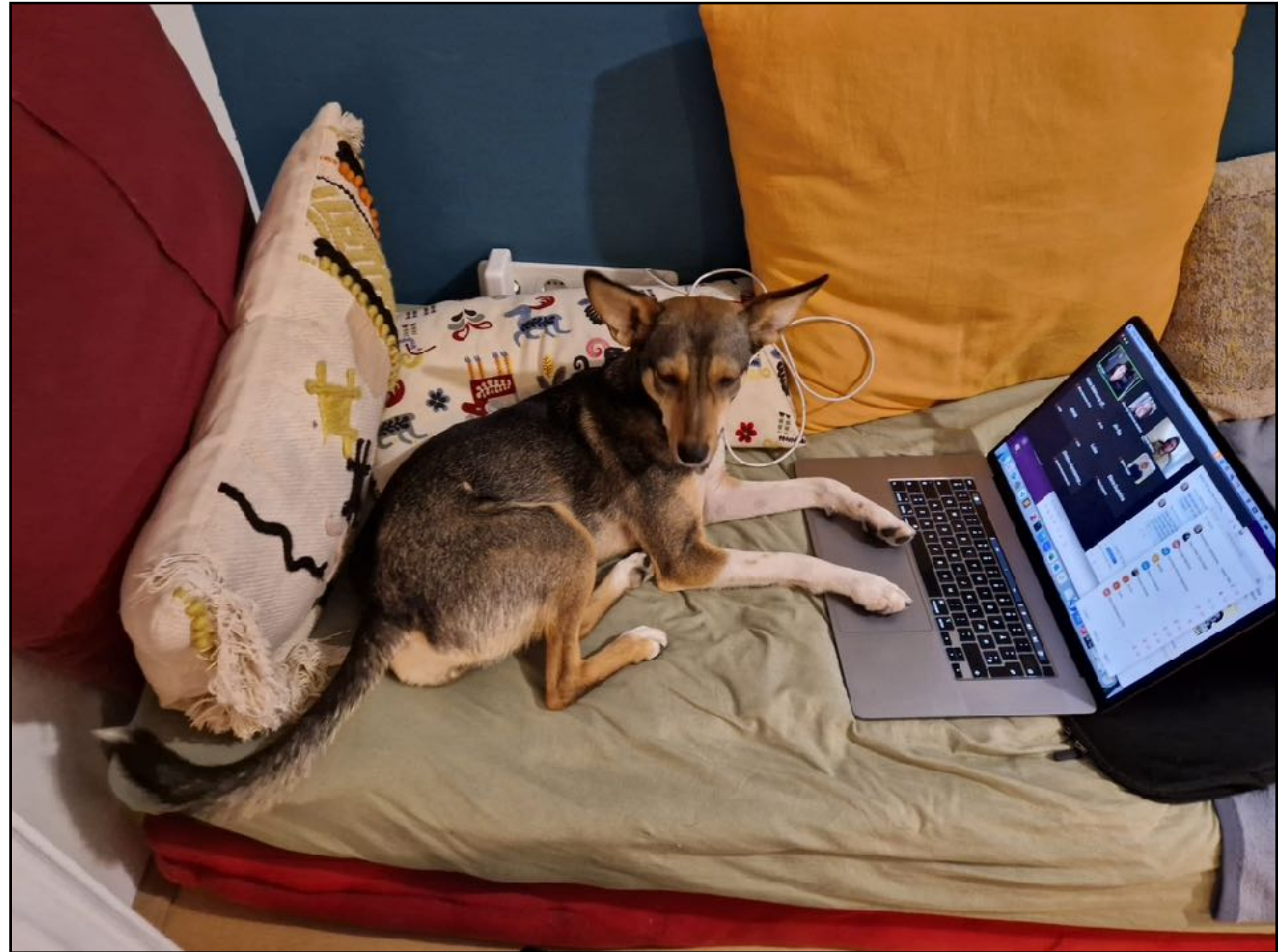
1. Open up your CLI. ((Windows: Anaconda Prompt; macOS/linux: Terminal))
2. Create a new conda environment, ``geotest``.
3. Activate the ``geotest`` environment.
4. Install, in the ``geotest`` environment, the following packages: ``geopandas``, ``jupyter``, ``ipykernel``.
5. Install kernelspecs for ``geotest`` by running:

```
python -m ipykernel install --user --name geotest
```

6. Run the lecture notebook with the ``geotest`` environment (+ change kernel!) – do you get the Copenhagen street network plot?

**Next steps for your programming career
(aka propaganda from Anastassia):**

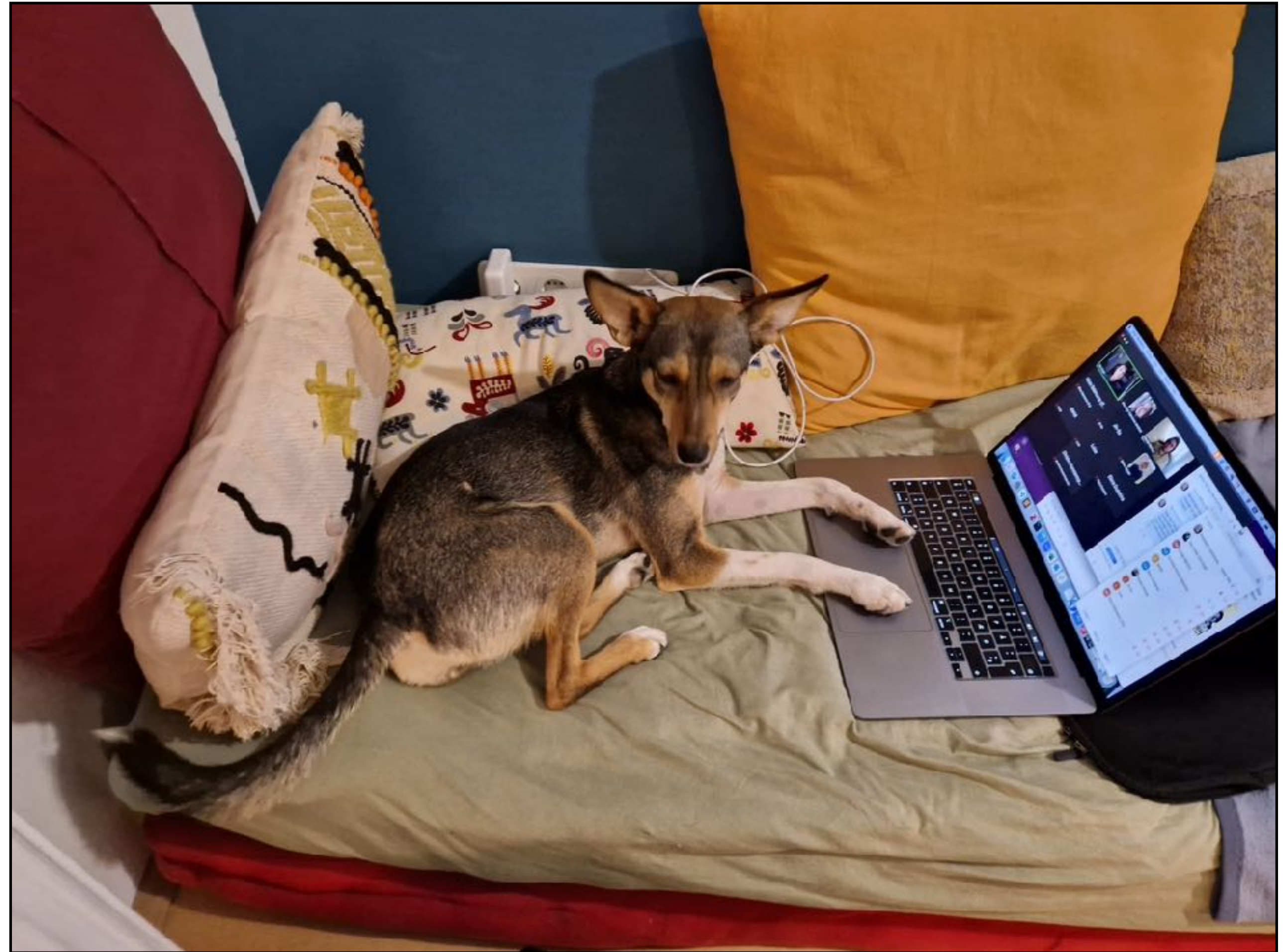
- **Continue learning**
- **The missing semester**
- **Version control & GitHub**
- **Open-source software - become a contributor**
- **Thu, October 30: guest lecture by Martin Fleischmann**



- You now already know: Python, markdown, regex, html, bash, (a tiny little bit of SQL) - continue learning!
- Take a class on The Missing Semester of Your Computer Science Education: <https://missing.csail.mit.edu/>
- Get familiar with version control with git & get familiar with GitHub. For example, take this short online course: <https://swcarpentry.github.io/git-novice/>
- Become a contributor to open-source software. For example, for pandas: <https://pandas.pydata.org/docs/development/contributing.html>
- Make sure to attend the lecture on Thu, October 30 for some more programming inspiration - guest lecture by [Martin Fleischmann](#)

[not just for
programming, but still
useful!]

- **Overleaf & Zotero**
- **WAIT & WAIT;
WAIC & WAINC**
- **BSc thesis creativity**



[not just for programming, but still useful!]

- For your student papers, use Overleaf (LaTeX) & Zotero (automated bibliography management)
- For your group projects, do not default to what you know best. In any group situation you can ask yourself WAIT/WAINT (“why am I talking” or “why am I not talking”); and also: WAIC/WAINC (“why am I coding” or “why am I not coding”).
- For your bachelor projects, start planning early and consider the sky the limit

Part 2: Recap

Operators

Mathematical

+ - * / ** // %

Assignment

= += -= *= /= **= %=

Logical

> < >= <= == != and or not

Bitwise logical

& |

Functions

Built-in

type()
isinstance()
print()
id()
round()
input()
open()
len()
all()
any()
sorted()

Built-in methods

.copy()
.apply()

Built-in set/list/tuple methods

.add()
.remove()
.pop()
.count()
.index()
.sort()
.reverse()

Built-in dict methods

.keys()
.values()
.items()

CLI commands

	windows	macOS/linux
Print out current working directory	chdir	pwd
Print out files in current working directory	dir	ls
Change working directory	cd <folderpath>	
Run python script	python <filepath>	
Open Python interpreter in CLI	python	
Exit Python interpreter in CLI	exit() + ENTER	

Built-in modules

math
random
os
re
time
collections
operator
csv
json
requests

Python packages

pandas
matplotlib.pyplot
beautifulsoup4
numpy

Keywords

if elif else and
not or True
False def return
in assert for
while break del
raise pass
import as from
None with try
except lambda

Coding environment & coding concepts

jupyter notebook interface

Markdown language

#comments in code

assigning a value to a variable

concatenating strings

calling a function

indexing & slicing (for str, list)

control flow

conditional statements

for loops while loops

assert statements

try/except (catching errors)

defining & calling your own functions

variable scope: global & local

arguments & parameters (required vs. optional)

string formatting f'{}' r'' \n \t

regular expressions

GUI (graphical user interface) vs. CLI (command line interface)

absolute vs. relative file paths

“Magic” (% for inline and %% for cell magic) in Jupyter notebook

%run <filepath> for running a Python file in Jupyter notebook

inline functions with lambda

list comprehension

type checking

random seeding

%%time and %%timeit for timing Jupyter notebook cell runtime

01/01/1970: Unix time starts

Conda for environment management

Monte Carlo simulations (random sampling)

Writing your own modules (.py files with function definitions)

HTML & webscraping

pandas functions

```
pd.read_csv()  
pd.DataFrame()  
pd.concat()
```

pandas attributes (for DataFrames)

```
.index  
.column  
.shape
```

pandas methods (for DataFrames)

```
.head()  
.tail()  
.describe()  
.sort_values()  
.drop()  
.reset_index()  
.unique()  
.isna()  
.notna()  
.fillna()  
.replace()  
.groupby()  
.copy()
```

pandas operators

```
~ (tilde - inversion)
```

**(cf Jupyter notebook
lecture11.ipynb)**



💜 Thank you



💜 Thank you



😎 Bring a pair of (black) sunglasses